

THE LOCAL WHITTAKER FACTOR AT A SPLIT LEVEL PRIME, AND AN INCOMPLETENESS IN THE STANDARD RECIPE FOR SCHOER'S

κ

ABSTRACT. Schofer's formula expresses averages of $\log\|\Psi_F\|$ over CM points on a Shimura curve in terms of quantities $\kappa_\eta(m)$ built from derivatives of Fourier coefficients of an incoherent Eisenstein series. The recipe used in practice to evaluate $\kappa_\eta(m)$ descends from Kudla–Rapoport–Yang, through Schofer and Errthum, to Yang; its central step enumerates the configurations that contribute to the derivative at the centre. We observe that this enumeration is derived under a restriction which Kudla–Rapoport–Yang state explicitly — that the Eisenstein series be built from sections given by the characteristic functions of the completions of $\mathcal{O}_k$ — and which is not satisfied by the lattices arising for an Eichler order of squarefree level N at a *split* level prime.

We compute the local Whittaker factor of the relevant N -scaled binary lattice at the level prime in closed form for every valuation, and show that its value at the critical point is $(N-1)\text{ord}_N(m)$. We then re-run the case analysis with this factor in place and find that it produces no additional contributing configuration, so that no $\log N$ can arise from $\kappa_\eta(m)$ for $m > 0$. Independently, a numerical evaluation of the vector-valued input form determines, exactly, a correction of the form $A_m \log N$ that the CM values demonstrably require. The rule producing it — half the constant term of the input form at a nonzero isotropic coset — is verified against ground truth on four bases with $N \in \{2, 3, 5\}$, including a base with $N = 5$ where the ground truth is forced by an over-determined consistency relation measured independently of any Eisenstein computation. We identify the correction as the constant-term pairing of the input form with the Eisenstein series attached to a nonzero isotropic coset of the discriminant form — an object the maximal-order literature does not compute — and reduce it, by an oldform relation of Schwagenscheidt, to ordinary seed-0 data of an enlarged lattice whose level-prime part is unimodular; this explains structurally both the support rule and the observed normalisation. We record the exact data any corrected formula must reproduce, and the routes we have eliminated.

1. INTRODUCTION

Let B be an indefinite quaternion algebra over $\mathbb{Q}$ of reduced discriminant D , let $\mathcal{O} \subset B$ be an Eichler order of squarefree level N with $(N, D) = 1$, and let $L = \{\alpha \in \mathcal{O} : \text{tr } \alpha = 0\}$ be the associated lattice of signature $(1, 2)$, so that $L^\vee/L \cong \mathbb{Z}/2 \times \mathbb{Z}/DN \times \mathbb{Z}/DN$ has order $2(DN)^2$ [4, Lemma 8]. For a weakly holomorphic vector-valued modular form F of weight $1/2$ and type ρ_L , Borchers' lift Ψ_F is a meromorphic modular form on $X_0^D(N)/W_{D,N}$, and Schofer's formula [7, 4] evaluates

$$\sum_{\tau \in \text{CM}(d)} \log|\psi_F(\tau)| = -\frac{|\text{CM}(d)|}{4} \sum_{\eta \in L^\vee/L} \sum_{m \geq 0} c_\eta(-m) \kappa_\eta(m),$$

valid under the hypotheses $O_{L,F}^+ = O_L^+$, $c_0(0) = 0$ and $c_\eta(m) \in \mathbb{Z}$ for $m \leq 0$.

The quantities $\kappa_\eta(m)$ are not given in [4]; the reader is referred to [3] and [9] for their evaluation. Both go back to the computation of Kudla–Rapoport–Yang [5], whose Proposition 2.8 identifies the configurations contributing to the derivative of an incoherent Eisenstein series at the centre and asserts that *these are the only cases which contribute*.

What this note contains. Section 2 traces that assertion to its stated scope. Section 3 computes the missing local ingredient: the local Whittaker factor of the N -scaled binary lattice at a split level prime, in closed form (Theorem 3.1), with the corollary that the empirical “level support rule” of the computational literature is exactly the vanishing of that factor (Corollary 3.2). Section 4 re-runs the case analysis with this factor and finds no new case (Proposition 4.1). Section 5 records the correction that the CM values require, computed exactly, and Section 6 lists the explanations we have eliminated. Section 8 states what remains open.

All numerical statements below are exact rational computations, carried out in Magma; the supporting scripts are described in Appendix A.

2. THE CHAIN, AND THE SCOPE OF ITS CENTRAL STEP

Write $U = \lambda^\perp$ for the negative definite plane attached to a CM point of discriminant d , $L_+ = L \cap \mathbb{Q}\lambda$ and $L_- = L \cap U$. Following [9, (11),(12)], one sets

$$\kappa_\eta(m) = \sum_{\mu \in L/(L_+ + L_-)} \sum_{x \in \eta_+ + \mu_+ + L_+} \kappa_{\eta_- + \mu_-}^-(m - \langle x, x \rangle/2),$$

where $\kappa_\mu^-(m)$ is the s -linear coefficient of the m -th Fourier coefficient $A_\mu(s, m, v)$ of an incoherent Eisenstein series of weight 1 attached to L_- , and $\kappa_\mu^-(0) = 0$ unless $\mu = 0$.

For $m > 0$ one has, with $S_{m,\mu} = \{p : p \mid \Delta \text{ or } v_p(m) > 0\}$ and $\Delta = \text{disc}(L_-)$,

$$A_\mu(s, m, v) = -\frac{2\pi}{L(s+1, \chi_d)} \prod_{p \in S_{m,\mu}} \frac{W_{m,p}(s, \varphi_{\mu,p})}{1 - \chi_d(p)p^{-1-s}}, \quad (1)$$

a genuine Euler product because $\varphi_\mu = \text{char}(\mu + \widehat{L_-})$ is a pure tensor. Since $A_\mu(s, m, v)$ vanishes at $s = 0$, some factor must vanish there, and differentiating (1) yields [9, Lemma 16]: the derivative is the single term at the vanishing place, and vanishes outright if two or more factors vanish.

That step is correct as an identity. What it inherits from [5] is the identification of the vanishing of a local factor with a *representability* obstruction, i.e. with membership of $\text{Diff}(m)$. Kudla–Rapoport–Yang state the scope of their computation plainly:

“in section 2, we have considered only those Eisenstein series built from sections which are defined by the characteristic functions of the completions of $\mathcal{O}_k$. More general sections could be considered at the cost of (i) more elaborate calculations of Whittaker functions and their derivatives used in computing the Fourier expansion of the central derivative, and (ii) incorporation of a level structure in the moduli problem.” [5, §6]

(They also fix a maximal order $\mathcal{O}_B$.) The lattice L_- arising here is not of that shape. Write $\Delta = \text{disc}(L_-)$. For a discriminant d with $N \nmid d$ — the case in which the level prime is *unramified* in $k = \mathbb{Q}(\sqrt{d})$, and hence, by Eichler’s condition on optimal embeddings, necessarily *split* — one has $\text{ord}_N(\Delta) = 2$: the binary lattice is N -scaled at the level prime, not unimodular. This is exactly a “more general section”.

Remark 2.1. The distinction is invisible in the maximal-order setting: there L_- is unimodular at every split prime, no split prime can obstruct representability, and correspondingly the logarithms produced by [7, Theorem 4.1] run over ramified and inert places only. Every discriminant admitting a CM point on $X_0^D(N)$ with $N \nmid d$ has N split, so this configuration is precisely the one the maximal-order theory never meets.

3. THE LOCAL WHITTAKER FACTOR AT A SPLIT LEVEL PRIME

Part (i) of the programme quoted above, at the level prime, is the following computation. Let $X = N^{-s}$.

Theorem 3.1. *Let $d < 0$ be a discriminant with $N \nmid d$ and let L_- be the negative definite binary lattice attached to a CM point of discriminant d , so $\text{ord}_N(\text{disc } L_-) = 2$. Then for every $m \geq 1$, with $j = \text{ord}_N(m)$,*

$$W_{m,N}(X) = 1 + (N-1)(X + X^2 + \cdots + X^j) - X^{j+1}.$$

In particular

$$W_{m,N}(1) = (N-1) \text{ord}_N(m), \quad W'_{m,N}(1) = (j+1) \left(\frac{(N-1)j}{2} - 1 \right).$$

The statement is a computation rather than a proof: it has been verified in exact arithmetic for $N = 2, 3, 5$ and all $1 \leq m \leq 60$, on the lattices L_- of $X_0^{15}(2)$, $X_0^6(5)$ and $X_0^{10}(3)$ at firing discriminants, giving 180 independent checks. It is pinned as a regression test in the accompanying code.

Corollary 3.2. *$W_{m,N}(1) = 0$ if and only if $N \nmid m$.*

This identifies, and explains, an empirical rule that has been observed repeatedly in computations with these curves: the correction discussed in Section 5 is supported exactly on the indices m with $N \mid m$. That rule is the vanishing of the level-prime factor.

Remark 3.3. At $j = 0$ the factor is $W_{m,N}(X) = 1 - X = 1 - N^{-s}$, the inverse local zeta factor. Its vanishing at $s = 0$ is therefore of a different nature from a representability obstruction: it holds identically for all m with $N \nmid m$, and at a split N the local quadratic space represents every m .

Remark 3.4. The value $W_{m,N}(1) = (N-1) \text{ord}_N(m)$ coincides with the local factor Kudla–Yang obtain at $p \mid N$ for the *quaternary* Eichler lattice [6, (9.1)]. The agreement appears to be a genuine coincidence of shape between the quaternary lattice and the binary L_- ; we do not have an explanation for it.

4. RE-RUNNING THE CASE ANALYSIS

With Theorem 3.1 in hand one can redo the enumeration of [5, Prop. 2.8] for the N -scaled section. Normalising as in (1), put $f_p(s) = W_{m,p}(s)/(1 - \chi_d(p)p^{-1-s})$.

Proposition 4.1. *At a firing discriminant, f_N has a simple zero at $s = 0$ exactly when $N \nmid m$, and this zero is not a Diff zero. Consequently*

$$\text{ord}_{s=0} A_\mu(s, m, v) = |\text{Diff}(m)| + [N \nmid m],$$

and since incoherence forces $|\text{Diff}(m)| \geq 1$:

- (1) *if $N \mid m$, the order is $|\text{Diff}(m)|$, and the derivative is nonzero precisely when $|\text{Diff}(m)| = 1$, contributing a logarithm at the ramified or inert place of $\text{Diff}(m)$;*
- (2) *if $N \nmid m$, the order is at least 2 and the derivative vanishes.*

In neither case does a log N occur.

So the level structure only ever raises the order of vanishing: it removes contributions, and creates no new contributing configuration. The three cases of [5, Prop. 2.8] stand, with no fourth. This is confirmed computationally: over roughly 190 terms (μ, x) arising in the evaluation of $\kappa_0(m)$ on three bases, the level prime is *never* the unique vanishing place, while it lies among two or more in 60–80% of terms.

Corollary 4.2. *Within this framework, a log N can enter a CM value only through the $m = 0$ term $c_0(0)\kappa_0(0)$, where $\kappa_0(0)$ carries the summand $\sum_{p|N/(N,d)} \log p$ of [9, Remark 15(2), Lemma 20].*

5. THE CORRECTION THE CM VALUES REQUIRE

Against this stands a computation of the input form itself. Following [4, eq. (6)] one may form $F_f = \sum_{\gamma \in \Gamma_0(M) \backslash SL_2(\mathbb{Z})} (f|_{1/2}\gamma) \rho(\gamma^{-1})e_0$ and evaluate it numerically; the metaplectic phase is handled by evaluating both the slash and ρ along one and the same ST -word, so that they share a single metaplectic lift. Every such evaluation is gated on reproducing the principal part of F_f predicted by [4, Lemma 24] from the two scalar q -expansions of f ; on the runs reported here that gate is met to 10^{-103} .

Observation 5.1. For every Borcherds form produced by the pipeline on $X_0^{15}(2)$, $X_0^6(5)$, $X_0^{10}(3)$ and $X_0^{21}(2)$ one finds $c_0(0) = 0$ (to 10^{-103}), so the hypothesis $c_0(0) = 0$ of [4, Thm. B] is satisfied and, by Corollary 4.2, no log N can arise.

Observation 5.2. The isotropic cosets of $L^\vee/L$ number exactly $2N - 1$; the D -part of the discriminant form is anisotropic, so all of them come from the level- N hyperbolic plane. All $2N - 2$ nonzero isotropic cosets carry the same constant term $c_\eta(0)$.

Observation 5.3. Let $\text{mult}(f) = \frac{1}{2}c_\eta(0)$ for any nonzero isotropic η . Then adding $\text{mult}(f) \cdot \sum_{p|N/(N,d_{\text{fund}})} \log p$ to the computed CM value reproduces, exactly, the corrections independently measured by kernel-consistency on $X_0^{15}(2)$ (9 values), $X_0^6(5)$ (5) and $X_0^{10}(3)$ (5), and by the consistency relation of Observation 5.4 below on $X_0^{14}(5)$ (1, at eleven digits): 20 of 20. A further base $X_0^{21}(2)$ supplies nine values with no prior ground truth against which to check them.

That the correction is genuinely required, and not an artefact of a normalisation, is visible on $X_0^6(19)$: on a set of CM points including discriminants divisible by N , the uncorrected values are mutually *inconsistent* — the associated system has no solution — whereas with the correction the published model is recovered.

Ground truth without an Eisenstein computation. The pipeline imposes, at every rational CM point d , the consistency relation

$$\varepsilon_1(d) \frac{s(d)}{s(d_A)} + \varepsilon_2(d) \frac{\tilde{s}(d)}{\tilde{s}(d_B)} = 1, \quad \varepsilon_i(d) \in \{\pm 1\}, \quad (2)$$

between the two Hauptmodul rows, normalised at the discriminants d_A, d_B where they vanish. Restoring a correction $r_f \log N$ to $\log|\psi_f|$ multiplies the normalised value at d by $N^{r_f(k(d) - k(d_{\text{ref}}))}$, where $k(d) \in \{0, 1\}$ records whether the term fires at d . Each rational CM point whose firing status differs from the reference's therefore contributes one equation in (N^{r_1}, N^{r_2}) ; two determine the pair and any further points are consistency checks. This measures the multipliers of the two Hauptmodul forms on any base the pipeline reaches, in seconds, with no Eisenstein series evaluated.

Observation 5.4. On nine bases the relation (2) determines the pair (r_{-1}, r_{-2}) uniquely:

$$\begin{array}{llll} X_0^{15}(2): (4, 2) & X_0^{14}(5): (1, 1) & X_0^6(7): (1, 2) & X_0^6(5): (0, 0) \\ X_0^{10}(3): (0, 0) & X_0^{10}(11): (0, 0) & X_0^6(19): (0, 0) & X_0^{22}(7): (0, 0) \\ X_0^{34}(5): (0, 0) & & & \end{array}$$

Every value with an independent determination is reproduced — $X_0^{15}(2)$ against the kernel-consistency measurement, $X_0^6(5)$ and $X_0^{10}(3)$ against the numerical evaluation of F_f , and $X_0^6(7)$,

the one *nonzero* cross-check, against an earlier independent measurement. Conversely, on $X_0^{14}(5)$ the pair $(1, 1)$ was measured first by (2) and *then* confirmed by the numerical evaluation of F_f : $\frac{1}{2}c_\eta(0) = 1.000000000000$ on the form with pole order 16, with the principal-part gate met to $2.3 \cdot 10^{-37}$ and all eight nonzero isotropic cosets agreeing. This is the first verification of the rule of Observation 5.3 at $N > 2$ against ground truth obtained by an entirely independent method.

Writing $\text{mult}(f) = \sum_m c(-m)A_m$ and using Borchers’ obstruction to set $A_m = -b(m)/4$ with b the relevant weight $3/2$ Eisenstein coefficient, the linear functional is determined per base by the measured multipliers. Without further hypotheses the system leaves four free directions on each base; imposing the support rule of Corollary 3.2 (the weight vanishes unless $N \mid m$) removes them all, and the resulting weights are *unique* with six spare conditions satisfied on each of the three bases — eighteen independent chances to falsify the support rule, none taken. Table 1 records the result.

base	A_m (equivalently $b(m) = -4A_m$)		
$X_0^{15}(2)$	$A_2 = -1$	$A_{10} = 1$	$A_{30} = 0$
$X_0^6(5)$	$A_{10} = 3/2$	$A_{15} = 1/2$	$A_{30} = 3/2$
$X_0^{10}(3)$	$A_3 = 1/2$	$A_{12} = 1/2$	$A_{30} = 3/2$
$X_0^{21}(2)$	$A_2 = -2, A_6 = -4 \quad A_{18} = 2, A_{42} = -8 \quad (0\text{-block: } B_{21} = -5)$		

TABLE 1. The correction, per index, as a linear functional of the principal part. The values are d -independent: at every firing discriminant the coefficient of $\log N$ in $\kappa_0(m)$ must equal A_m , whereas the recipe of §2 returns 0.

6. EXPLANATIONS ELIMINATED

Each of the following was tested against the data of Section 5 and is inconsistent with it. We record them because several are natural first guesses.

- (1) $\text{mult} = c_0(0)$, or any linear combination of the two scalar constant terms $\text{Coeff}(f_\infty, 0)$ and $\text{Coeff}(f_0, 0)$: inconsistent on three bases.
- (2) Evaluating the local Whittaker at a nonzero isotropic coset η^* rather than at 0: destroys the pattern of vanishing indices on all three bases. The mechanism is now clear: the nonzero isotropic cosets live in the level- N hyperbolic plane, so changing the coset changes *only* the level-prime factor (measured: from $1 - X^2/N$ to the constant 1, with the factors at $p \mid D$ byte-identical) and can never repair an index whose defect is at a ramified prime.
- (3) Replacing the local factor at $p \mid D$ by the anisotropic branch of [6, Prop. 5.3]: does not reproduce b , under either sign of κm . (The implementation of [6, Props. 5.3, 5.4] agrees with the pipeline’s own factor at every good prime, 470 checks, and at the level prime, 150 checks.)
- (4) The incoherent support condition $|\text{Diff}(m)| = 1$: scores 8/9 on the exact coefficients and is the only rule that predicts $b(30) = 0$ on $X_0^{15}(2)$, but is far too permissive — for $r \leq 60$ it holds at 15, 19 and 20 indices with $N \nmid r$, where the coefficient must vanish.
- (5) Cancelling the zeta factor $1 - X$ at the level prime before counting vanishing places: the recovered terms differentiate at ramified and inert primes, never at N .
- (6) The inner $m = 0$ of κ_η , i.e. terms with $\langle x, x \rangle/2 = m$: never occurs, in all 60 (discriminant, index) pairs over every firing discriminant of the three bases.

- (7) Schofer's outer $m = 0$ term: vanishes by Observation 5.1.
- (8) The re-enumeration of §4: adds no case.
- (9) Any re-weighting of the existing local data: every combination of a subset of the prefactors $\{\sqrt{m}/\sqrt{|d|}, h/w, \prod(1 - \chi(p)/p), \prod(1 - 1/p^2)^{-1}\}$ with a local rule drawn from {the product of the $W_p(1)$; the derivative of that product; the derivative taken at N , at the primes $p \mid D$, at each single prime, or at the vanishing places}, times an arbitrary constant per base, fails to reproduce the nine measured multipliers of $X_0^{15}(2)$.
- (10) More strongly, *no multiplicative correction of any kind* can succeed: at two of the nine determined indices (both with $m = 10$) the product of the code's local factors vanishes while the true weight does not, so the discrepancy is infinite there. The missing contribution is a *term*, not a factor.
- (11) Differentiating at the place where the ternary *space* fails to represent m (Kudla's Diff, computed at the level of quadratic spaces rather than lattice densities): reproduces $A_2 = -1$ and $A_{30} = 0$ on $X_0^{15}(2)$ exactly — the only rule found that *explains* $A_{30} = 0$ — but fails on 7 of 9 indices, and is refuted as a support rule by $X_0^6(5)$ at $m = 15$, where Diff is empty but $A_{15} = 1/2 \neq 0$. The support is $N \mid m$; whether Diff supplies the correct place to differentiate is a separate question.

7. THE CORRECTION IDENTIFIED

The rule of Observation 5.3 reads off the constant term $c_{\eta^*}(0)$ of the input form at a nonzero isotropic coset η^* . By the constant-term pairing (pair F against a holomorphic Eisenstein series of the complementary weight $3/2$ and take the constant term of the resulting weight-2 invariant), that quantity is determined by the principal part of F paired against the coefficients of E_{A,η^*} , the Eisenstein series attached to the coset η^* of the discriminant form $A = L^\vee/L$. This is precisely the object the maximal-order literature does not compute: [2], [6, §8–9] and the recipe of §2 all work with the series attached to the *zero* coset (equivalently, with sections $\text{char}(\mu + \widehat{L})$ seeded at 0), and Kudla–Yang explicitly leave the general Eichler-order case open at the end of their §9.

The coefficients of $E_{A,\beta}$ for isotropic $\beta \neq 0$ are computed by Schwagenscheidt [8]. For our purposes the essential ingredient is not the main formula but the oldform relation [8, Prop. 1.4] applied to the trivial character: for $\beta = \eta^*$ of prime order $N_\beta = N$,

$$E_{A,\eta^*,\chi_0} = E_{B,0} \uparrow_B^A - E_{A,0}, \quad B = \langle \eta^* \rangle^\perp / \langle \eta^* \rangle \cong L_1^\vee / L_1, \quad (3)$$

where L_1 is the even lattice generated by L and η^* . Pairing F against (3) and using Observations 5.1 and 5.2 — $c_0(0) = 0$, the constant $c_\eta(0)$ shared by all $2N - 2$ nonzero isotropic cosets, and the vanishing of the constant-term pairing for every nontrivial character twist — gives

$$\text{mult}(f) = -\frac{1}{4(N-1)} \sum_{n>0} \sum_{\gamma \in (\eta^*)^\perp} c_\gamma(-n) b_{\gamma+H}^{B,0}(n), \quad H = \langle \eta^* \rangle, \quad (4)$$

in which every object on the right is a *seed-0* Eisenstein coefficient of the smaller module B — the territory of [2], computable by the same machinery as the existing recipe, applied to L_1 in place of L .

Three structural facts corroborate the identification.

- (1) The pairing constant in (4) is $4(N-1)$ — exactly the normalisation measured in Observation 5.3 (the sum of $c_\eta(0)$ over the $2N - 2$ nonzero isotropic cosets equals $4(N-1) \cdot \text{mult}$).
- (2) $\det L_1 = \det L / N^2$: the level-prime part of L_1 is *unimodular*, so the coefficients $b^{B,0}$ have no vanishing level-prime factor. The zero $W_{m,N}(1) = 0$ at $N \nmid m$ (Corollary 3.2),

which removes the majority of terms from the seed-0 pairing on A , never occurs on B — the structural mechanism by which (4) can produce the $\log N$ terms that §4 shows the seed-0 recipe cannot.

- (3) $2|\det L_1|$ remains a perfect square (e.g. $2 \cdot 450 = 30^2$ on $X_0^{15}(2)$), so the discriminant and character bookkeeping of the weight-3/2 coefficient formula carries over unchanged in shape.

Two cautions, the first of which the data has since promoted to a finding. The identity (3) is proven in [8] for weight $k \geq 5/2$; at $k = 3/2$ both sides require analytic continuation, and the naive pairing argument assumes the continued series is holomorphic at the special point, where Zagier-type non-holomorphic contributions are possible. That assumption *fails*, and it fails in a way one can measure without any of the machinery of this section: for the true seed-0 series on A itself, the constant-term pairing forces $\sum_{n>0,\gamma} c_\gamma(-n) b_\gamma^{A,0}(n)$ to equal $-c_0(0)$ minus a spread term proportional to the multiplier — so on a form with $\text{mult} = 0$ and $c_0(0) = 0$ it must vanish outright. Evaluating the left side with the naïve weight-3/2 coefficient formula (rank 3 collapses the Bruinier–Kuss local polynomials to plain representation densities, since they are evaluated where the factor $1 - p^{m-1}X$ vanishes) gives -5 , not 0, on such a form of $X_0^{15}(2)$. The naïve recipe therefore violates its own constant-term identity, independently of everything specific to (4) — and this, rather than any defect of (3), is where the computation currently stops. The discrepancy is exactly the shape of the harmonic completion: the ξ -image of the weight-3/2 Eisenstein series is a weight-1/2 theta on A (supported, by Serre–Stark, on the isotropic lines — the same level- N structure that carries everything else in this note), and its contribution to the pairing is, for weakly holomorphic input, a *finite* Weyl-vector-type functional of the principal part in the sense of [1, §10]. Calibrating that functional — for which the $\text{mult} = 0$ forms give equations free of every other unknown — is the immediate next step. We note the required care: the lattices here are anisotropic over $\mathbb{Q}$, so the unfolding of [1, §10] does not apply verbatim and the evaluation must proceed through the finite quadratic module.

8. THE OPEN QUESTION

Corollary 4.2 and Observation 5.1 are in direct tension with Observation 5.3 and with the failure of the uncorrected computation on $X_0^6(19)$. We see two possibilities and cannot presently distinguish them.

- (1) The constant term $c_0(0)$ of the form F actually attached to ψ_F differs from that of the coset sum F_f . The principal parts agree to 10^{-103} , but the principal part alone does not determine $c_0(0)$.
- (2) The correction compensates an error elsewhere in the computation of the CM values, outside Schofer’s formula. It is a d -independent linear functional of the principal part, which is the shape of a per-form normalisation.

The identification of §7 sharpens the second reading. The $m = 0$ term of the formula, as transmitted through [7, 9], collapses the sum $\sum_\eta c_\eta(0) \kappa_\eta(0)$ to the single term $c_0(0) \kappa_0(0)$ via a condition of the form $\delta_{0,\mu}$. But the collapse condition constrains only the *negative-definite* part of the coset, so a nonzero isotropic η with $\eta_- = 0$ is not excluded by it — and by Observations 5.2 and 5.3 it is exactly those cosets, with their shared nonzero constant term, that carry the correction. On this reading nothing outside Schofer’s formula is being compensated; rather, terms inside it were dropped.

In either case Theorem 3.1 and Proposition 4.1 stand, and part (ii) of the Kudla–Rapoport–Yang programme — the incorporation of a level structure — remains to be carried out. Table 1 is what any corrected formula must reproduce, index by index.

A third reading should be recorded, because it is not excluded by anything above. The evaluation of $\kappa_\mu^-(m)$ for $m > 0$ has never been checked against an independent computation at $N > 1$ — neither here nor, as far as we can determine, in the literature. Yang’s worked examples [9, Ex. 21–24] take $B = (\frac{-1,3}{\mathbb{Q}})$ with a *maximal* order, and indeed $L = L_+ + L_-$ there; the corresponding regression tests in the accompanying code likewise cover only $N = 1$. What Observation 5.4 now supplies is a verification at $N > 1$ of the *correction rule* — the multiplier $\frac{1}{2}c_\eta(0)$, confirmed at $N = 5$ against ground truth measured by an independent method — but the evaluation of $\kappa_\mu^-(m)$ itself, which sits upstream in the $m > 0$ terms, remains unverified at $N > 1$.

Remark 8.1 (Published CM values need not discriminate). A natural-seeming check is to compare against a published CM value on a base with $N > 1$ and a nonvanishing correction; $X_0^{14}(5)$ with $s(\tau_{-280}) = 5/16$ [4, Ex. 36] appears to be the only available instance. It cannot work. The Hauptmodul there is normalised by $s(\tau_{-4}) = \infty$, $s(\tau_{-11}) = 1$, $s(\tau_{-35}) = 0$, so the published value depends only on the ratio $s(-280)/s(-35)$ — and -35 and -280 are both non-firing ($5 \mid d$), so any correction multiplies both by the same power of N and cancels; the remaining anchors sit at 0 and ∞ , where a multiplicative factor is invisible. Indeed the pipeline reproduces $5/16$ exactly both with and without the correction. A published value can only test the correction if its normalising anchors and the tested point lie on opposite sides of the firing rule; this should be checked, at a cost of nothing, before any such comparison is attempted. The consistency relation (2), by contrast, is sensitive exactly where [4, Ex. 36] is blind, and is what supplied the $X_0^{14}(5)$ ground truth here.

APPENDIX A. REPRODUCIBILITY

The computations were carried out in Magma against the codebase accompanying this note.

- **VectorValuedForm.m** constructs F_f : coset representatives for $\Gamma_0(M) \backslash SL_2(\mathbb{Z})$, ST -words, the weight $1/2$ metaplectic slash, the Weil representation over $\mathbb{C}$, and the Fourier extraction. The action of $\rho(S)$ is implemented as a finite Fourier transform on $L^\vee/L$, which reduces the cost from $|G|^2$ to $|G| \sum_r d_r$ and the memory from $|G|^2$ to $|G|$; without this the case $|G| = 24200$ is out of reach.
- **EisensteinLocalFactors.m** implements [6, (2.16), (2.17), Props. 5.3, 5.4] and exposes **LocalWhittakerPolynomial**, the Whittaker polynomial whose s -derivative produces the logarithms.
- Numerical parameters matter. The horocycle height must satisfy $\Im \tau \geq 1$, or some coset is carried into the pole at ∞ and the terms overwhelm their sum. The number of sample points must beat the positive Fourier coefficients, which grow like $\exp(4\pi\sqrt{p\overline{m}})$ for pole order p : on the pole-order-30 forms of $X_0^{15}(2)$ the principal-part gate reads $5.8 \cdot 10^{73}$, $1.2 \cdot 10^{62}$, $2.7 \cdot 10^{28}$ and $8.7 \cdot 10^{-103}$ at $K = 48, 64, 96$ and 192 respectively. Raising the working precision without raising K has no effect.
- Every reported value is gated on the principal-part check; a run failing that gate is discarded.

REFERENCES

- [1] R. E. Borcherds, *Automorphic forms with singularities on Grassmannians*, Invent. Math. **132** (1998), 491–562.
- [2] J. H. Bruinier and M. Kuss, *Eisenstein series attached to lattices and modular forms on orthogonal groups*, Manuscripta Math. **106** (2001), 443–459.
- [3] E. Errthum, *Singular moduli of Shimura curves*, Canad. J. Math. **63** (2011), 826–861.

- [4] J.-W. Guo and Y. Yang, *Equations of hyperelliptic Shimura curves*, Proc. Lond. Math. Soc. (2017), doi:10.1112/S0010437X16007739.
- [5] S. Kudla, M. Rapoport and T. Yang, *On the derivative of an Eisenstein series of weight one*, Internat. Math. Res. Notices (1999), no. 7, 347–385.
- [6] S. Kudla and T. Yang, *Eisenstein series for $SL(2)$* , Sci. China Math. **53** (2010), 2275–2316.
- [7] J. Schofer, *Borcherds forms and generalizations of singular moduli*, J. reine angew. Math. **629** (2009), 1–36.
- [8] M. Schwagenscheidt, *Eisenstein series for the Weil representation*, J. Number Theory **193** (2018), 74–101.
- [9] Y. Yang, *Special values of hypergeometric functions and periods of CM elliptic curves*, arXiv:1503.07971.